

MADai

Enhancement and Integration Ideas

A strategic backlog for expanding MADai and connecting it to the MADProspects Universe.

Document	Scope	Version
Enhancement and Integration Ideas	MADai application, API, workers, operations, and MADProspects ecosystem	Prepared 2026-05-26

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1. Strategic Direction

MADai should become the autonomous operations layer of MADProspects: a shared control plane where every MAD application can request work, publish events, receive generated artifacts, and expose operational health. The strongest opportunity is not only adding screens to MADai, but making MADai the queue, worker, workflow, and self-healing nervous system for the broader product universe.

North-star idea

Every MADProspects product should be able to send a task to MADai, receive status updates, attach context, receive artifacts, and trigger follow-up actions without each product inventing its own worker system.

1.1 Product Principles

- Use MADai as the cross-product task and automation backbone.
- Keep each application domain-specific, but standardize identity, eventing, health, task execution, and observability.
- Expose humans to simple workflows while allowing workers and integrations to use richer API contracts.
- Prefer configurable playbooks and templates over hard-coded one-off automations.
- Treat every task, artifact, webhook, deployment, and recommendation as auditable.

2. High-Impact Enhancements Inside MADai

2.1 Full Operator Dashboard

Build the restored frontend into a complete operations cockpit. The dashboard should show queue health, live workers, task throughput, failed tasks, dead-letter backlog, average duration, success rate, audit findings, webhook failures, and deployment status. It should support drill-down from metric to affected task, worker, company, endpoint, or audit finding.

2.2 Task Board and Task Detail Views

- Kanban-style columns for Draft, Queued, Scheduled, Claimed, Running, AwaitingValidation, Completed, Failed, DeadLetter, and Cancelled.
- Task detail with prompt payload, input payload, output summary, result payload, validation report, logs, comments, artifacts, retries, dependencies, and execution history.
- Bulk actions for retry, cancel, assign queue, change priority, add tags, and export.
- Saved filters for product, company, queue, category, status, priority, owner, worker, and created date.
- Timeline view showing status changes, logs, comments, and artifacts in order.

2.3 Worker Fleet Management

- Live worker cards with status, queue, max concurrency, current concurrency, last heartbeat, CPU, memory, disk, labels, and capabilities.
- Drain, deactivate, maintenance mode, and restart-request workflows.

- Worker capability matching so tasks can require skills like Angular, .NET, SQL, PDF generation, image generation, scraping, or deployment.
- Worker performance scoring by success rate, average duration, category, and failure pattern.
- Workspace inspection showing recent logs, artifacts, and claimed tasks.

2.4 Claude Task Board

Turn /app/claude into a real SystemAdmin board for repository-improvement work. Add columns for Pending, InProgress, ToBeDeployed, Completed, Failed, Cancelled, and Deferred. Support attachments, prompt templates, priority, notes, scanner import review, deployment grouping, and verification checklists.

2.5 Template and Playbook Library

Create reusable playbooks that generate one or more tasks from a template. Examples: migrate one app to madprospects.com, deploy frontend and API, add superadmin seed, create PDF documentation, recover missing published files, audit an app folder, or standardize pnpm store usage.

2.6 Human Approval Gates

Add structured approvals before high-risk actions: deploy, database migration, mass email, credential rotation, domain change, destructive cleanup, billing change, or tenant-wide setting changes. Approval records should be auditable and tied to the task execution timeline.

2.7 Artifact Gallery

Create an artifact area where task outputs are browsable by product, company, task, type, and date. This can store PDFs, DOCX files, screenshots, deployment logs, test reports, generated images, HTML exports, and data extracts.

3. Intelligence and Automation Ideas

3.1 Universal Work Intake

Add an intake form where users describe what they want in plain language. MADai classifies the request into category, priority, product, required worker skills, risk level, likely subtasks, and suggested template. It then creates a structured task or multi-task workflow.

3.2 AI Planning Layer

Before execution, an AI planner can break work into research, design, implementation, test, deploy, and documentation phases. The plan should be editable by a human and saved as task dependencies.

3.3 Automated QA and Evidence Packs

Every task that changes code or production should produce an evidence pack: build output, test output, health checks, screenshots, deployed URLs, changed files, and rollback notes. Evidence packs become artifacts and can be attached to release notes.

3.4 Self-Healing Operations

- Detect repeated failures by endpoint, worker, product, category, or exception type.
- Auto-create diagnostic tasks with logs and affected entity links.
- Suggest fixes and confidence levels.
- Allow an operator to approve self-healing execution.
- Close the loop by rechecking health after the fix.

3.5 Prompt Template Governance

Version prompt templates. Track which template produced which task result. Add ratings, success metrics, rollback capability, and environment-specific template variants.

3.6 Multi-Agent Execution

Introduce specialized worker roles: planner, coder, reviewer, tester, deployer, documenter, researcher, data analyst, and compliance checker. Workflows can route through multiple agents with explicit handoff artifacts.

3.7 Cost and Capacity Intelligence

Use the billing and usage domains to track estimated and actual effort, worker runtime, token usage where available, artifact storage, and queue wait time. Surface capacity forecasts and recommend when to scale workers.

4. MADProspects Universe Integration Map

Application	What it does	How MADai can integrate
MADLeads	AI-first lead management, SaaS subscriptions, feature flags, AI providers, quota management.	Create lead-enrichment, campaign-generation, CRM cleanup, data-quality, and AI-cost-analysis tasks. Receive webhook events for new leads, failed syncs, quota alerts, and campaign milestones.

Application	What it does	How MADai can integrate
MADHub	Marketing and operations cockpit for entrepreneurs and founders.	Use MADai as the execution backend for marketing playbooks, content calendars, operational checklists, and cross-app status dashboards.
MADCreate	Website and app generation platform.	Route generation, editing, deployment, DNS checks, site audits, screenshot tests, copywriting, and repair tasks through MADai workers.
MADAuthor	AI-native book creation and publishing platform.	Run research, outline, chapter drafting, editing, formatting, export, cover, KDP metadata, and marketing asset pipelines as task workflows.
MADRecruiting	Recruitment platform with local API/frontend targets.	Create candidate screening, job post generation, interview pack, reference check, CRM follow-up, and hiring pipeline tasks.
MADPulse	Venue discovery, ratings, and promotions.	Run venue data enrichment, scraping, moderation, promotion generation, geospatial checks, review summarization, and venue onboarding tasks.
MADLearn	Learning management for South African businesses.	Generate courses, quizzes, learning paths, certificates, transcript analysis, assessment reports, and compliance reminders.
MADLove	Relationship operating system.	Create guided reflection packs, health summaries, conflict-repair playbooks, reminders, and anonymized pattern analysis with careful privacy controls.
MADCloud	Repository and product orchestration dashboard.	Share repository intelligence, health checks, deployment orchestration, worker fleet status, and cross-product automation with MADai.
MADHeroes	Leads and venue inquiry platform for Heroes Advertising.	Turn venue inquiries, newsletter signups, and lead forms into response, qualification, follow-up, and reporting tasks.
Multisciple	Path-based multi-tenant church management SaaS.	Automate tenant onboarding, member import, event schedules, sermon content, communications, audit checks, and custom-domain verification.

5. Cross-Product Integration Patterns

5.1 Shared Event Bus Through Webhooks

Each MAD application can emit events into MADai: lead.created, site.deploy.failed, book.chapter.ready, candidate.shortlisted, venue.promotion.expiring, course.completed, tenant.created, or health.failed. MADai stores the event, creates tasks where appropriate, and delivers signed callbacks when work progresses.

5.2 Shared Task API

Every product should be able to create a MADai task using a stable contract: product key, tenant/company, category, title, description, priority, queue, due date, input payload, artifact URLs, callback endpoint, and approval policy.

5.3 Shared Identity and Role Mapping

Longer term, MADai can participate in a MADProspects identity layer. Users should not maintain separate accounts per product. Roles can map to product scopes, while SystemAdmin and support roles remain ecosystem-wide.

5.4 Shared Artifact Storage

Task artifacts should be reusable across products. A MADAuthor chapter export, MADCreate screenshot, MADLeads CSV, or MADRecruiting interview pack should have metadata, owner, retention policy, and source task links.

5.5 Shared Audit Language

Standardize audit findings across apps: severity, status, product, tenant, source, evidence, recommendation, owner, due date, related tasks, and resolution notes. MADai then becomes the shared audit-remediation engine.

5.6 Shared Deployment Control Plane

MADai can coordinate deployments across API/frontend pairs. A deployment task can build, upload, inject environment values, hit health checks, verify browser login, take screenshots, and mark a release as passed or failed.

6. Product-Specific Playbook Ideas

6.1 MADLeads Playbooks

- Lead enrichment: append company size, industry, contact role, likely intent, and recommended next action.
- Campaign copy generation: produce email, SMS, WhatsApp, and landing-page variants from a lead segment.
- Quota alert remediation: analyze heavy AI usage and recommend cheaper provider/model choices.
- Stripe lifecycle monitor: create tasks for failed payments, expiring trials, and upgrade opportunities.

6.2 MADCreate Playbooks

- Generate SME website from brief: sitemap, copy, theme, imagery prompts, pages, SEO metadata, and deploy checklist.
- Site repair: crawl production, detect broken pages/assets, create fix tasks, and verify screenshots.

- Domain onboarding: DNS checklist, SSL check, tenant routing, and deployment proof.
- Conversion audit: review above-the-fold message, CTAs, page speed, accessibility, and lead capture.

6.3 MADAuthor Playbooks

- Book pipeline: concept, market scan, outline, chapter briefs, drafts, edits, continuity pass, export, and launch copy.
- Transcription to book: ingest sermons, voice notes, or interviews and convert to chapters.
- KDP preparation: metadata, categories, keywords, back-cover copy, author bio, and launch assets.
- Series continuity: maintain canon, terminology, voice, and structure across volumes.

6.4 MADRecruiting Playbooks

- Job advert generation from role brief and company profile.
- Candidate screening summary with skills, risks, questions, and ranking rationale.
- Interview pack generation for hiring managers.
- Reference-check follow-up automation and status tracking.
- Offer-letter drafting and onboarding checklist generation.

6.5 MADPulse Playbooks

- Venue onboarding: enrich profile, normalize location, validate categories, and suggest promotional copy.
- Review summarization: turn ratings and comments into insight cards for venue owners.
- Promotion optimizer: recommend time windows, target audiences, and offer wording.
- Quality monitor: flag stale venue data, suspicious reviews, and missing media.

6.6 MADLearn Playbooks

- Course generation from a company SOP, policy, or training need.
- Quiz and assessment generation with answer keys and remediation paths.
- Learner-risk monitor that creates follow-up tasks for incomplete or failed modules.
- Certificate and compliance pack generation for regulated training.

7. Technical Architecture Enhancements

7.1 Integration SDKs

Create TypeScript and .NET SDKs for creating tasks, sending events, reading status, downloading artifacts, and verifying webhook signatures. This keeps each product from hand-writing MADai integration code.

7.2 Product Registry

Add a Product table for MADLeads, MADHub, MADCreate, MADAuthor, MADRecruiting, MADPulse, MADLearn, MADLove, MADCloud, MADHeroes, Multisciple, and future products. Store URLs, API endpoints, health checks, owners, deployment targets, source paths, environment names, and integration capabilities.

7.3 Secure Secret Broker

Do not put product secrets in task payloads. Add references to secret names and resolve them only inside approved workers. Log secret access without logging secret values.

7.4 Workflow Engine

Introduce a workflow layer above tasks. A workflow defines phases, dependencies, approval gates, rollback steps, artifacts, SLAs, and notifications. Tasks remain the executable units.

7.5 Standard Health Contract

Every product should expose /health and /health/ready with a consistent JSON shape. MADai can poll them, show product status, and create remediation tasks on failure.

7.6 Browser Verification Workers

For frontend apps, a browser worker can log in, click critical flows, capture screenshots, compare visual regressions, and attach proof. This is especially useful after FTP/Plesk deployments.

7.7 Data Connectors

Add connectors for SQL Server, PostgreSQL, Stripe, SMTP, GitHub, Google Drive, SharePoint, Slack/Teams, WhatsApp providers, Google Analytics, Search Console, and domain/DNS providers. Each connector should be permissioned and auditable.

8. Governance, Security, and Compliance Ideas

8.1 Risk Scoring

Every task should have a risk score based on action type, environment, affected product, tenant count, data sensitivity, and required permissions. High-risk tasks should require approval and evidence packs.

8.2 Data Privacy Controls

- Classify task payloads as public, internal, confidential, personal, or sensitive.
- Redact secrets and personal data from logs by default.
- Add retention policies for artifacts, logs, and webhook events.

- Support tenant-specific privacy rules for products handling relationships, recruiting, learning, or church data.

8.3 Audit Trails

Every create, update, delete, status transition, approval, deployment, webhook delivery, secret access, and artifact download should be auditable. The existing auditable entity foundation is a strong start.

8.4 Human-in-the-Loop Defaults

MADai should automate aggressively but deploy conservatively. Risky tasks should stage recommendations, present evidence, and ask a SystemAdmin or product owner to approve execution.

9. Suggested Roadmap

Phase	Theme	Deliverables
Phase 1	Make current MADai fully usable	Full frontend: dashboard, task board, worker fleet, Claude board, admin basics, settings, files, notifications, webhooks.
Phase 2	Cross-product intake	Product registry, shared task API, SDKs, webhook event standard, MADProspects product health board.
Phase 3	Automation playbooks	Template library, workflow engine, approval gates, evidence packs, browser verification worker.
Phase 4	Self-healing universe	Shared audit language, product health monitors, auto-created remediation tasks, post-fix verification.
Phase 5	Commercial intelligence	Usage/cost dashboards, tenant billing hooks, capacity forecasts, AI provider optimization, SLA reporting.
Phase 6	Autonomous operations network	Multi-agent orchestration, product-specific specialist agents, cross-app campaign and delivery workflows.

9.1 Immediate Backlog

- 1 Build a real /app/claude board from the existing Claude tasks API.
- 2 Rebuild task list/detail/create screens and connect them to /api/v1/tasks.
- 3 Add worker fleet screens and live SignalR updates.
- 4 Add product registry seeded from Setup.xlsx and local MADProspects folders.
- 5 Add a universal task intake form with template suggestions.
- 6 Add frontend screens for audit findings, webhooks, notifications, settings, files, and admin.
- 7 Create integration SDKs for .NET and TypeScript.
- 8 Add a deployment playbook that builds, deploys, health-checks, and browser-verifies each app.

10. Success Metrics

Metric	Why it matters
Task cycle time	Shows how quickly requests move from intake to done.
Queue wait time	Shows whether worker capacity is sufficient.
Worker utilization	Shows idle, busy, and overloaded agents.
Success rate	Shows reliability by product, queue, category, and worker.
Dead-letter count	Shows unresolved failures requiring operator attention.
Automation acceptance rate	Shows whether AI suggestions are useful enough for operators to approve.
Deployment verification pass rate	Shows release quality.
Cross-product tasks created	Shows whether MADai is becoming the universe control plane.
Evidence completeness	Shows whether work is auditable and supportable.

Recommended first milestone

Make MADai the trusted operator cockpit for its own repository first, then connect one high-value product such as MADRecruiting or MADCreate through the shared task API. Prove the pattern with one complete workflow before rolling it out to the whole universe.